RUBÉN SERRANO NIETO, PhD

R&D RESEARCHER, MATERIALS SCIENTIST, CHEMIST

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PROFILE

PhD-trained Materials Scientist specialised in the design, synthesis, and extensive characterisation of functional hybrid materials, particularly Metal-Organic Frameworks (MOFs). Experienced in applied R&D, aiming to translate fundamental materials chemistry into technologies with real-world impact.

EMPLOYMENT

R&D Researcher at IMDEA Energy Institute, Mostoles (Madrid), Spain Mar 2022 — Dec 2025

• Led the design, synthesis, and optimisation of novel hybrid materials (>2,000 syntheses, 18 novel materials, 12 structures resolved) using high-throughput, solvothermal, and microwave-assisted synthetic approaches. • Conducted extensive multi-technique characterisation (up to 12), including structural, textural, and spectroscopic methods. • Designed and executed workflows linking structure–property–function, enabling selection of promising materials for energy and environmental applications (photocatalytic H_2 production, wastewater remediation, CO_2 conversion to cyclic carbonates, and polymerization catalysis) reaching benchmark-level efficiencies under mild conditions. • Collaborated in 6 competitive R&D projects (EU, national, regional; H2-MOF, MOFSEIDON, C-MOF.cell, NAPOLION, H+MOFs, VIRMOF-CM), contributing to data analysis AND technical reporting. • Authored 5 peer-reviewed publications in high-impact journals and presented at 12 scientific conferences, including international symposia • Held technical responsibility for laboratory instrumentation and unit inventory management.

Research assistant at IMDEA Energy Institute, Mostoles (Madrid), Spain Apr 2021 — Feb 2022

Synthesis of new MOFs with ionic conductive properties for their application as exchange membranes in fuel cells (PEMFCs).

EDUCATION

PhD in Advanced Chemistry, Complutense University of Madrid (UCM), Madrid (Spain) Nov 2022 — Dec 2025

PhD's thesis: "Novel Phosphonate-based Metal-Organic Frameworks (P-MOFs): From Design to Application", oriented to the synthesis and exhaustive characterisation of novel P-MOFs for applications in catalysis. Awarded Sobresaliente Cum Laude and International PhD Mention. Directed by Patricia Horcajada (h-index = 84; ~40,000 cites; >270 publications) at the Advanced Porous Materials Unit (APMU) of IMDEA Energy Institute.

MSc in Chemical Science & Technology, UCM, Madrid, Spain Sep 2020 — Sep 2021

Master's oriented to nanoscience and the material field. MSc's thesis: "New electro/photoactive MOFs: synthesis, characterization and properties", dealing with the synthesis of new MOFs with potential applications in catalysis (green hydrogen production) and water remediation (antibiotic elimination) (APMU, IMDEA Energy).

BSc in Chemistry, UCM, Madrid (Spain) Sep 2015 — Jul 2020

BSc's thesis: "Materials for clean energy storage based on porous metal-organic frameworks (MOFs)", focused on the synthesis of new MOFs with potential applications in energy storage as electrolytes (APMU, IMDEA Energy).

INTERNSHIPS

University of Glasgow (UofG), Scotland (UK) Jun 2023 — Oct 2023

Within the framework of the María de Maeztu Mobility Programme, solar cell prototypes were developed using bismuth-based perovskites and quantum-dot-MOFs at Pablo Docampo's group (h-index = 54; ~25,000 citations; 140 publications).

Instituto Fundación Teófilo Hernando I+D del Medicamento (IFTH) - UAM Jun 2019 — Oct 2019

Development and implementation of synthetic methods for preparing new organic molecules potentially active in the treatment of neurodegenerative diseases caused by the P2X7 receptor (1 peer-reviewed publication in a high-impact journal)

TECHNICAL SKILLS

- Spanish (native)
- English (C1)
- Mandarin (HSK 2, HSKK 1)
- Applied AI tools for research workflows
- Data processing (Origin)
- Mercury, Checkcell, OMNIC, FullProf, WinGX, Olex, DigitalMicrograph, TriStar II Plus
- Adobe Illustrator, Affinity, and Blender 3D
- Structural database analysis (CCDC-CSD)
- Microsoft Office (Word, Excel, PowerPoint)
- X-Ray Diffraction (SC/P-XRD)
- Crystal Refinement (LeBail)
- Sorption studies
- Scanning Electron Microscopy (SEM)
- Transmission Electron Microscopy (TEM)
- Thermogravimetric Analysis (TGA)
- Elemental analysis (EA)
- Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES)
- Infrared Spectroscopy (FTIR)
- UV-Vis Spectroscopy
- Nuclear Magnetic Resonance (NMR)
- High Performance Liquid Chromatography (HPLC)
- Organic, microwave (mono and multimodal), and solvothermal syntheses
- Shaping (pelletizing) and metallization
- Evaluation of the stability of materials under relevant conditions (pH, glovebox)
- Scientific writing and technical reporting

SCIENTIFIC PRODUCTION (ORCiD: 0000-0002-7785-0615)

- **'Ca-Triazine Phosphonate Metal-Organic Framework as Fungicide Active Packaging'.** Rubén Serrano-Nieto, Miriam Abán-Alfaro, Claire Ayala, Marta E. G. Mosquera, Tania Hidalgo, Inés Álvarez (Inorganic Chemistry Frontiers 2025, Accepted)
- **'Controlling Band-Bending for Perovskite Optoelectronic Devices Using Bismuth-Based Interlayers'.** Rubén Serrano-Nieto, Wai Kin Yiu, Marcin Giza, Fraser J. Angus, Graeme Cooke, Patricia Horcajada, Yolanda Pérez, Pablo Docampo (ACS Omega 2025, Accepted)
- **'Triazine Mg-Based Phosphonate Metal–Organic Framework for ε-Caprolactone Polymerization'.** Rubén Serrano-Nieto, Christian Rentero, Pablo Salcedo-Abraira, Catalina Biglione, Agustin M. Corner Manrique, Thomas Devic, Valentina Sessini, Marta E. G. Mosquera, Patricia Horcajada (Inorganic Chemistry 2025, Accepted)
- **'Novel pyrene phosphonate metal-organic frameworks to efficiently degrade sulfamethazine in wastewater'**. Rubén Serrano-Nieto, Sergio Carrasco, Samuel Morales-Cámara, Antonio Rodríguez-Diéguez, Catalina Biglione, Pablo Salcedo-Abraira, Yolanda Pérez, Sara Rojas, Patricia Horcajada (Journal of Environmental Chemical Engineering 2025, Accepted)
- **'Two New Cu-based Phosphonate Metal-Organic frameworks as Efficient Water Splitting Photocatalysts'**. P. Salcedo-Abraira, R. Serrano-Nieto, C. Biglione, M. Cabrero-Antonino, A. A. Babaryk, D. Tilve-Martínez, A. Rodríguez-Diéguez, S. Navalón, H. García, P. Horcajada (Chemistry of Materials 2023, Accepted)
- **'Novel Purine Derivative ITH15004 Facilitates Exocytosis through a Mitochondrial Calcium-Mediated Mechanism'**. Ricardo de Pascual, Francesco Calzaferri, Paula C. Gonzalo, Rubén Serrano-Nieto, Cristóbal de los Ríos, Antonio G. García, Luis Gandía (International Journal of Molecular Sciences 2022, Accepted)

SCIENTIFIC MEETINGS

- **Dec. 2024.** Oral communication at the **13th Annual Workshop of Young Researchers of IMDEA Energy.** IMDEA Energy, Spain (Oral, 2 days). 'Novel MOFs based on Phosphonate Ligands: Synthesis, Characterization and Application'. R. Serrano-Nieto, P. Salcedo-Abraira, S. Morales-Cámara, Catalina Biglione, S. Carrasco, M. Cabrero-Antonino, S. M.F. Vilela, A.A. Babaryk, D. Tilve-Martínez, A. Rodriguez-Diéguez, S. Navalón, Y. Pérez, S. Rojas, H. García, P. Horcajada
- **July 2024.** Oral communication at **VIII Simposio Anual en Química Avanzada (SAQA).** UCM, Spain (Oral, 1 day). 'Novel Metal-Organic Frameworks based on Phosphonate Ligands: Synthesis, Characterization and Application'. R. Serrano-Nieto, P. Salcedo-Abraira, S. Morales-Cámara, Catalina Biglione, S. Carrasco, M. Cabrero-Antonino, S. M.F. Vilela, A.A. Babaryk, D. Tilve-Martínez, A. Rodriguez-Diéguez, S. Navalón, Y. Pérez, S. Rojas, H. García, P. Horcajada
- **June 2024.** International oral communication at the **International Workshop on Layered and Nanostructured Materials (IWLNM24).** UPO, Perugia, Italy (Oral, 4 days). 'Pyrene Phosphonate Metal-Organic Frameworks for Environmental Remediation '. R. Serrano-Nieto, S. Morales-Cámara, C. Biglione, S. Carrasco, A. Rodriguez-Diéguez, P. Salcedo-Abraira, Y. Pérez, S. Rojas, P. Horcajada
- **June 2024.** Oral communication at **XX Reunión Bienal del Grupo Especializado de Química Inorgánica y XIV Reunión Bienal del Grupo Especializado de Química del Estado Sólido (QIES24).** UDC, A Coruña, Spain (Oral, 3 days). 'Pyrene Phosphonate Metal-Organic Frameworks in the Removal of Contaminants of Emerging Concern'. R. Serrano-Nieto, S. Morales-Cámara, C. Biglione, S. Carrasco, A. Rodriguez-Diéguez, P. Salcedo-Abraira, Y. Pérez, S. Rojas, P. Horcajada
- **Sep. 2023.** International poster communication at **5th European Conference on Metal Organic-Frameworks and Porous Polymers (EUROMOF23).** UGR, Granada, Spain (Poster, 2 days). 'PokéMOF: Catalyse with 'em all'. S. Carrasco, P. Salcedo-Abraira, R. Serrano-Nieto, R. M. Guerrero, I. Rincón Lucas, A.J. Chacón-García, J. Ferrando-Ferrero, C. Fernández-Ruiz, I.D. Lemir, C. Biglione, T. Hidalgo, Y. Pérez, P. Horcajada.
- **May 2023.** Oral presentation at **IV Congreso sobre Materiales Multifuncionales (MultiMat IV).** Almuñécar, Granada, Spain (Oral presentation, 2 days). "Two novel Cu-based phosphonate MOFs as efficient water splitting photocatalysts". R. Serrano-Nieto, P. Salcedo-Abraira, C. Biglione, M. Cabrero-Antonino, S. M.F. Vilela, A. A. Babaryk, D. Tilve-Martínez, A. Rodriguez-Diéguez, S. Navalón, H. García, P. Horcajada.
- **April 2023.** Oral presentation at **VII Jornada de promoción de la Investigación Básica para Estudiantes de Ciencias e Ingenierías.** URJC, Madrid, Spain (oral flash communication, 2 days). "Novel Cu & Bi pyrene phosphonate MOFs as promising materials for energy related applications". R. Serrano-Nieto, P. Salcedo-Abraira, C. Biglione, N. Ureña, T. Perez-Prior, F. Salles, A. Várez, M. Cabrero-Antonino, S. Navalón, S. M.F. Vilela, A. A. Babaryk, D. Tilve-Martínez, A. Rodriguez-Diéguez, H. García, P. Horcajada.
- **June 2022.** Attendance to **XVI Congreso Nacional de Materiales (CNMAT2022).** UCLM, Castilla-La Mancha, Spain (Poster, 4 days). "Nuevos MOFs como electrolitos en PEMFCs". R. Serrano-Nieto, P. Salcedo-Abraira, C. Biglione, N. Ureña, T. Perez-Prior, F. Salles, A. Várez, P. Horcajada. Awarded with best poster presentation
- **June 2022.** Oral presentation at **VI Jornada de promoción de la Investigación Básica para Estudiantes de Ciencias e Ingenierías.** URJC, Madrid, Spain (oral flash communication, 2 days). "Nuevos MOFs como materiales prometedores en dispositivos de generación de energía limpia". R. Serrano-Nieto, P. Salcedo-Abraira, C. Biglione, N. Ureña, T. Pérez-Prior, F. Salles, A. Várez, P. Horcajada.
- **Dec. 2021.** Attendance at the **10th Annual Workshop of Young Researchers of IMDEA Energy.** IMDEA Energy, Madrid, Spain (Poster, 2 days). **'**Novel pyrene-based MOFs: synthesis, characterization and properties'. R. Serrano-Nieto, P. Salcedo-Abraira, C. Biglione, S. Morales-Cámara, S. Rojas, A. Rodríguez, S. Navalón, P. Horcajada.

- **Sept. 2021.** Oral presentation at **V Jornada de promoción de la Investigación Básica para Estudiantes de Ciencias e Ingenierías.** URJC, Madrid, Spain (oral flash communication, 2 days). "Synthesis and characterization of MOFs with potential applications as photocatalysts". R. Serrano-Nieto, P. Salcedo-Abraira, C. Biglione, A. Rodríguez, S. Navalón, P. Horcajada.

COURSES

- **Jan 2025.** Attendance at the course '**Iesmat Seminar | Gas Adsorption in Porous Materials'.** IMDEA Energy, Madrid, Spain (Course, 8 hours).
- **June 2024.** Attendance at the course '**Graphic design applied to scientific research'.** UCM, Madrid, Spain (Course, 16 hours).
- **Feb 2024.** Attendance at the course '**Application of Artificial Intelligence to PhD thesis for students'.** UCM, Madrid, Spain (Course, 15 hours).
- **Feb 2024.** Attendance at the course **'Blender: Simulation and 3D Science'.** UCM, Madrid, Spain (Course, 16 hours).
- **Feb-May 2023.** Attendance at the course **'Experimental techniques in Chemistry'.** UCM, Madrid, Spain (Course, 34 hours).
- **Feb. 2023.** Attendance at the course **'Advanced Training on Gas Sorption'** from Anton Paar Spain S.L.U. ICMM-CSIC, Madrid, Spain (Online, 3 hours). Speaker: Martin Thomas
- **Nov. 2022.** Attendance at the course **'Introduction to Electrochemical Processes'.** IMDEA Energy, Madrid, Spain (Course, 2 days).
- **June 2022.** Attendance at the **Workshop on 'Innovative methods for nanoMOF synthesis and characterization'** from the HeatNMof training network. IMDEA Energy, Madrid, Spain (Online, 32 hours). Coordinator: Patricia Horcajada.
- **Feb. 2022** Attendance at **XIV Workshop on Structural Resolution Using X-ray Diffraction.** Almuñécar, Granada, Spain (Course, 20 hours).
- **Jan. 2022** Attendance at the course **'Synthetic methods, characterisation and applications of porous nanostructured materials'.** URJC, Madrid, Spain (Course, 20 hours).
- **July 2021.** Attendance at the **Workshop of 'Dissemination of research results and ethics'** from the HeatNMof training network. IMDEA Energy, Madrid, Spain (Online, 21 hours). Coordinator: Patricia Horcajada.

DISSEMINATION

Newspaper articles

- **March 2025.** News article at **The Conversation**. “Los antibióticos están llegando a nuestros grifos: ¿podemos evitarlo?”. Rubén Serrano Nieto, Patricia Horcajada y Yolanda Pérez Cortés. https://theconversation.com/los-antibioticos-estan-llegando-a-nuestros-grifos-podemos-evitarlo-250297
- **April 2024.** News article at **The Conversation**. “Así mejoramos las pilas de hidrógeno para que sean más duraderas y asequibles”. Rubén Serrano Nieto, Catalina Biglione y Patricia Horcajada. https://theconversation.com/asi-mejoramos-las-pilas-de-hidrogeno-para-que-sean-mas-duraderas-y-asequibles-224576

Scientific workshops

- **'Science Week'** (2023-2024). IMDEA Energy, Spain.
- **'Día de la Mujer y la Niña en la ciencia'** (2023-2024). IMDEA Energy, Spain.
- **'Researchers Night'** (2023-2024). IMDEA Energy, Spain